Q1. JEE Main 2026 (21 January Shift 2)

Given below are two statements:

Statement I : The correct order in terms of bond dissociation enthalpy is $Cl_2 > Br_2 > F_2 > I_2$.

Statement II: The correct trend in the covalent character of the metal halides is $[SnCl_4 > SnCl_2]$, $[PbCl_4 > PbCl_2]$ and $[UF_4 > UF_6]$.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

Q2. JEE Main 2026 (21 January Shift 2)

The correct increasing order of $C-H(A), C-O(B), C=O(C)$ and $C \equiv N(D)$ bonds in terms of covalent bond length is:

(1) $D < C < A < B$

(2) $A < D < C < B$

(3) $D < C < B < A$

(4) $A < B < C < D$

Q3. JEE Main 2026 (23 January Shift 1)

Identify the molecule (X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among HNO_3, H_2SO_4, NF_3 and O_3. Choose the correct bond angle made by the central atom of the molecule (X).

(1) 116°

(2) 102°

(3) 107°

(4) 120°

Q4. JEE Main 2026 (28 January Shift 2)

Match List - I with List - II according to shape.

	List-I		List-II
$A.$	XeO_3	$I.$	BrF_5
$B.$	XeF_2	$II.$	NH_3
$C.$	XeO_2F_2	$III.$	$[I_3]^-$
$D.$	$XeOF_4$	$IV.$	SF_4

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-II, C-IV, D-I

Q5. JEE Main 2026 (24 January Shift 2)

Pair of species among the following having same bond order as well as paramagnetic character will be-

(1) O_2^-, N_2^+

(2) O_2^+, N_2^{2-}

(3) O_2^-, N_2^-

(4) O_2^+, N_2^-

Q6. JEE Main 2026 (22 January Shift 2)

Among H_2S, H_2O, NF_3, NH_3 and $CHCl_3$, identify the molecule (X) with lowest dipole moment value. The number of lone pairs of electrons present on the central atom of the molecule (X) is :

(1) 0 (2) 3 (3) 2 (4) 1

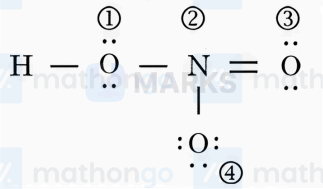
Q7. JEE Main 2026 (22 January Shift 1)

Two p-block elements X and Y form fluorides of the type EF_3. The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

(1) sp^2 and sp^3
(2) Both sp^3
(3) sp^3 and sp^2
(4) Both sp^2

Q8. JEE Main 2026 (22 January Shift 1)

The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



(1) $0, 0, -1, +1$
(2) $0, -1, 0, +1$
(3) $+1, 0, 0, -1$
(4) $0, +1, 0, -1$

Q9. JEE Main 2026 (23 January Shift 2)

Which statements are NOT TRUE about XeO_2F_2 ?

A. It has a see-saw shape.
B. Xe has 5 electron pairs in its valence shell in XeO_2F_2.
C. The $O - Xe - O$ bond angle is close to $180°$.
D. The $F - Xe - F$ bond angle is close to $180°$.
E. Xe has 16 valence electrons in XeO_2F_2.

Choose the correct answer from the options given below:

(1) B, C and E Only
(2) B and D Only
(3) A and D Only
(4) B, D and E Only

Q10. JEE Main 2026 (21 January Shift 1)

Given below are two statements :

Statement I : The number of species among $SF_4, NH_4^+, [NiCl_4]^{2-}, XeF_4, [PtCl_4]^{2-}, SeF_4$ and $[Ni(CN)_4]^{2-}$, that have tetrahedral geometry is 3.

Statement II : In the set $[NO_2, BeH_2, BF_3, AlCl_3]$, all the molecules have incomplete octet around central atom.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

Q11. JEE Main 2026 (28 January Shift 1)

Given below are two statements:

Statement I: The number of species among BF_4^-, SiF_4, XeF_4 and SF_4, that have unequal $E-F$ bond lengths is two. Here, E is the central atom.

Statement II: Among O_2^-, O_2^{2-}, F_2 and O_2^+, O_2^- has the highest bond order.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are false
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are true

Q12. JEE Main 2026 (24 January Shift 1)

Among the following, the CORRECT combinations are

A. $IF_3 \rightarrow$ T-shaped $(sp^3 d)$
B. $IF_5 \rightarrow$ Square pyramidal $(sp^3 d^2)$
C. $IF_7 \rightarrow$ Pentagonal bipyramidal $(sp^3 d^3)$
D. $ClO_4^- \rightarrow$ Square planar $(sp^2 d)$

Choose the correct answer from the options given below:

(1) B, C and D Only
(2) A and B Only
(3) A, B and C Only
(4) A, B, C and D

Q13. JEE Main 2026 (24 January Shift 1)

Given below are statements about some molecules/ions.

Identify the CORRECT statements.

A. The dipole moment value of NF_3 is higher than that of NH_3.

B. The dipole moment value of BeH_2 is zero.

C. The bond order of ${O_2}^{2-}$ and F_2 is same.

D. The formal charge on the central oxygen atom of ozone is -1.

E. In NO_2, all the three atoms satisfy the octet rule, hence it is very stable.

Choose the correct answer from the options given below:

(1) A, C & D Only
(2) B, C & D Only
(3) B & C Only
(4) A, B, C, D & E

ANSWERS AND SOLUTIONS

1. (3)	**2.** (2)	**3.** (2)	**4.** (3)	**5.** (4)	**6.** (4)	**7.** (1)	**8.** (4)
9. (1)	**10.** (2)	**11.** (1)	**12.** (3)	**13.** (3)			

1. **(3)** Statement I: The bond dissociation enthalpy of halogens follows the order $Cl_2 > Br_2 > F_2 > I_2$. The bond energy of F_2 is lower than Cl_2 and Br_2 due to the high inter-electronic repulsion between the lone pairs on the small fluorine atoms. Thus, Statement I is true.

Statement II: According to Fajan's rules, for a given metal, the compound with the metal in a higher oxidation state has more covalent character due to higher polarizing power. Therefore, $SnCl_4 > SnCl_2$ and $PbCl_4 > PbCl_2$ are correct. However, for Uranium halides, UF_6 has a higher oxidation state (+6) than UF_4 (+4), so UF_6 is more covalent than UF_4. The statement claims $UF_4 > UF_6$, which is incorrect. Thus, Statement II is false.

2. **(2)** Bond lengths depend on bond multiplicity and atomic sizes.
C-H (single) ≈ 1.09 Å, C-O (single) ≈ 1.43 Å, C=O (double) ≈ 1.23 Å, C≡N (triple) ≈ 1.16 Å.
Triple bonds are shorter than double bonds, which are shorter than single bonds.
For single bonds, C-O is longer due to lower electronegativity difference compared to C-H.
Increasing order of bond length: C-H(A) < C ≡ N(D) < C = O(C) < C-O(B), giving A < D < C < B.

3. **(2)** Analyzing lone pairs on central atoms:
HNO_3: N is central with 3 bonds and 0 lone pairs
H_2SO_4: S is central with 4 bonds and 0 lone pairs
NF_3: N is central with 3 bonds and 1 lone pair
O_3: Central O with 2 bonding regions and 1 lone pair
NF_3 has maximum lone pairs (1) with the clearest structure. N undergoes sp^3 hybridization with 1 lone pair and 3 bonding pairs giving trigonal pyramidal geometry. The F-N-F bond angle is reduced from ideal tetrahedral 109.5° due to lone pair repulsion, resulting in approximately 102°.

4. **(3)** Determine molecular shapes using VSEPR theory:
A. XeO_3 (3 bonds, 2 lone pairs on Xe): pyramidal
B. XeF_2 (2 bonds, 3 lone pairs on Xe): linear
C. XeO_2F_2 (4 bonds, 2 lone pairs on Xe): see-saw
D. $XeOF_4$ (5 bonds, 1 lone pair on Xe): square pyramidal
Matching with shapes:
I. BrF_5: square pyramidal → D-I
II. NH_3: pyramidal → A-II

III. $[I_3]^-$: linear → B-III

IV. SF_4: see-saw → C-IV

5. **(4)** Bond orders and magnetic character:

O_2^- (BO=1.5, paramagnetic), O_2^+ (BO=2.5, paramagnetic), O_2^{2-} (BO=1, diamagnetic);

N_2^+ (BO=2.5, paramagnetic), N_2^- (BO=2.5, paramagnetic), N_2^{2-} (BO=2, diamagnetic)

1: BO 1.5 vs 2.5 (different);

2: BO 2.5 vs 2 (different);

3: BO 1.5 vs 2.5 (different);

4: O_2^+ (BO=2.5, paramagnetic) and N_2^- (BO=2.5, paramagnetic) - both same bond order and paramagnetic character

6. **(4)** Comparing dipole moments:

$H_2S \approx 0.97$ D (bent, 2 lone pairs on S),

$H_2O \approx 1.85$ D (bent, 2 lone pairs on O),

$NF_3 \approx 0.24$ D (pyramidal, 1 lone pair on N),

$NH_3 \approx 1.42$ D (pyramidal, 1 lone pair on N),

$CHCl_3 \approx 1.04$ D (tetrahedral, 0 lone pairs on C).

NF_3 has the lowest dipole moment because the highly polar N-F bonds are oriented in such a way that their contributions nearly cancel with the effect of the lone pair. The central atom in NF_3 is N, which has 1 lone pair of electrons.

7. **(1)** XF_3 is a Lewis acid, indicating the central atom X lacks a complete octet and can accept electron pairs. This is characteristic of Boron (B).

Boron in BF_3:

Valence electrons: 3

Steric number: 3 (three B-F bonds, no lone pairs)

Geometry: Trigonal planar

Hybridization: sp^2

YF_3 is a Lewis base, indicating the central atom Y has lone pair(s) available to donate. This is characteristic of Nitrogen (N).

Nitrogen in NF_3:

Valence electrons: 5

Steric number: 4 (three N-F bonds + one lone pair)

Geometry: Trigonal pyramidal

Hybridization: sp^3

Therefore, the hybridizations are sp^2 for XF_3 (Boron) and sp^3 for YF_3 (Nitrogen).

8. **(4)** The formal charge (F.C.) of an atom in a Lewis structure is calculated using the formula:

F.C. = [Total valence electrons] - [Total non-bonding (lone pair) electrons] - $\frac{1}{2}$[Total bonding (shared) electrons]

For atom (1) Oxygen: Valence $e^- = 6$, Lone pair $e^- = 4$, Bonding $e^- = 4$.
$F.C. = 6 - 4 - \frac{1}{2}(4) = 0$
For atom (2) Nitrogen: Valence $e^- = 5$, Lone pair $e^- = 0$, Bonding $e^- = 8$.
$F.C. = 5 - 0 - \frac{1}{2}(8) = +1$
For atom (3) Oxygen: Valence $e^- = 6$, Lone pair $e^- = 4$, Bonding $e^- = 4$.
$F.C. = 6 - 4 - \frac{1}{2}(4) = 0$
For atom (4) Oxygen: Valence $e^- = 6$, Lone pair $e^- = 6$, Bonding $e^- = 2$.
$F.C. = 6 - 6 - \frac{1}{2}(2) = -1$
The formal charges are $0, +1, 0, -1$ respectively.

9. **(1)** In XeO_2F_2: Xe has 8 valence electrons forming 4 bonding pairs (2 with O, 2 with F) and 2 lone pairs, giving a trigonal bipyramidal electron geometry with a see-saw molecular shape, making statement A TRUE.
Statement B is NOT TRUE because the total electron pairs around Xe is 6 (4 bonding + 2 lone), not 5.
For statement C, the O-Xe-O angle in a see-saw geometry is approximately 120°, not close to 180°, making it NOT TRUE.
Statement D is TRUE as the F-Xe-F angle is approximately 180° when F atoms occupy axial positions.
Statement E is NOT TRUE since Xe has only 8 valence electrons, not 16.
The incorrect statements are B, C, and E.

10. **(2)** Statement I: Checking geometry of each species:
SF_4: 4 bp + 1 lp → see-saw
NH_4^+: 4 bp → tetrahedral
$[NiCl_4]^{2-}$: Ni^{2+} (d^8) with weak field Cl^- → sp^3 → tetrahedral
XeF_4: 4 bp + 2 lp → square planar
$[PtCl_4]^{2-}$: Pt^{2+} (d^8) → dsp^2 → square planar
SeF_4: 4 bp + 1 lp → see-saw
$[Ni(CN)_4]^{2-}$: Ni^{2+} with strong field CN^- → dsp^2 → square planar
Only 2 species are tetrahedral. Statement I is false.
Statement II: NO_2 (odd electron), BeH_2 ($4e^-$), BF_3 ($6e^-$), $AlCl_3$ ($6e^-$) all have incomplete octet. Statement II is true.

11. **(1)** Statement I: Let's analyze the geometry of the given species.
BF_4^-: sp^3 hybridization, tetrahedral geometry. All B-F bond lengths are equal.
SiF_4: sp^3 hybridization, tetrahedral geometry. All Si-F bond lengths are equal.
XeF_4: sp^3d^2 hybridization, square planar geometry. All Xe-F bond lengths are equal.
SF_4: sp^3d hybridization, see-saw geometry. It has axial and equatorial bonds. Axial bonds are longer than equatorial bonds due to greater repulsion. Thus, only SF_4 has unequal bond lengths. The number of species is one, not two. Statement I is false.
Statement II: Bond order calculation using Molecular Orbital Theory:

Total electrons in O_2^- = 17, Bond Order = 1.5
Total electrons in O_2^{2-} = 18, Bond Order = 1.0
Total electrons in F_2 = 18, Bond Order = 1.0
Total electrons in O_2^+ = 15, Bond Order = 2.5
Among these, O_2^+ has the highest bond order (2.5), not O_2^-. Statement II is false.
Both Statement I and Statement II are false.

12. (3) Analyzing geometry and hybridization:

A. IF_3: Iodine has 7 valence electrons, forms 3 bonds with F, leaving 2 lone pairs. Total electron pairs = 5 (trigonal bipyramidal geometry). Molecular shape is T-shaped. Hybridization = sp^3d. CORRECT.

B. IF_5: Iodine forms 5 bonds with F and has 1 lone pair. Total electron pairs = 6 (octahedral geometry). Molecular shape is square pyramidal with 1 lone pair in axial position. Hybridization = sp^3d^2. CORRECT.

C. IF_7: Iodine forms 7 bonds with F and has 0 lone pairs. Total electron pairs = 7. Molecular shape is pentagonal bipyramidal. Hybridization = sp^3d^3. CORRECT.

D. ClO_4^-: Chlorine has 7 valence electrons + 1 from charge = 8. Bonded to 4 O atoms with 0 lone pairs. Total electron pairs = 4 (tetrahedral geometry). Molecular shape is tetrahedral, NOT square planar. Hybridization = sp^3, NOT sp^2d. INCORRECT.

Correct combinations: A, B, and C.

13. (3) Statement A: In NH_3, the orbital dipole due to the lone pair is in the same direction as the resultant dipole moment of the $N-H$ bonds. In NF_3, the orbital dipole is in the opposite direction to the resultant dipole moment of the $N-F$ bonds. Thus, dipole moment of NH_3 (1.47 D) is higher than NF_3 (0.23 D). Statement A is incorrect.
Statement B: BeH_2 is a linear molecule (sp hybridization). The two $Be-H$ bond dipoles are equal and opposite, so they cancel each other out, making the net dipole moment zero. Statement B is correct.
Statement C: O_2^{2-} has 18 electrons. Its bond order is $\frac{10-8}{2}=1$. F_2 also has 18 electrons and its bond order is 1. Statement C is correct.
Statement D: In the Lewis structure of ozone (O_3), the central oxygen atom forms one double bond and one single bond and has one lone pair. Formal charge = $V-L-\frac{B}{2}=6-2-\frac{6}{2}=+1$. Statement D is incorrect.
Statement E: NO_2 is an odd-electron molecule (17 valence electrons). The nitrogen atom has only 7 electrons in its valence shell, so it does not satisfy the octet rule. Statement E is incorrect.
Therefore, only statements B and C are correct.